

Continuous measurements of flow rate in a shallow gravel-bed river by a new acoustic system

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[1] The continuous measurement of river discharge for long periods of time is crucial in water resource studies. However, the accurate estimation of river discharge is a difficult and labor-intensive procedure; thus, a robust and efficient method of measurement is required. Continuous measurements of flowrate have been carried out in a wide, shallow gravel bed river (water depth ≈ 0.6 m under low-flow conditions, width ≈ 115 m) using Fluvial Acoustic Tomography System (FATS) that has 25 kHz broadband transducers with horizontally omnidirectional and vertically hemispherical beam patterns. Reciprocal sound transmissions were performed between the two acoustic stations located diagonally on both sides of the river. The horizontal distance between the transducers was 301.96 m. FATS enabled the measurement of the depth- and range-averaged sound speed and flow velocity along the ray path. In contrast to traditional point/transect measurements of discharge, in a fraction of a second, FATS covers the entire cross section of river in a single measurement. The flow rates measured by FATS were compared to those estimated by moving boat Acoustic Doppler Current Profiler (ADCP) and rating curve (RC) methods. FATS estimates were in good agreement with ADCP estimates over a range of 20 to 65 $\text{m}^3 \text{s}^{-1}$. The RMS of residual between the two measurements was 2.41 $\text{m}^3 \text{s}^{-1}$. On the other hand the flowrate by RC method fairly agreed with FATS estimates for greater discharges than around 40 $\text{m}^3 \text{s}^{-1}$. This inconsistency arises from biased RC estimates in low flows. Thus, the flow rates derived from FATS could be considered reliable.

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1. Introduction

[2] The acquisition of continuous measurements of river discharge is a high-priority issue for water resource management, and the real-time knowledge of river discharge is an environmental, social, and economic asset. Therefore, establishing a method and/or technology for quantifying discharge is of paramount importance.

[3] One of the most widely practiced methods is to determine the stage-discharge relationship (rating curve) using many field observations. The rating curve method presents several limitations. *Di Baldassarre and Montanari* [2009] estimated rating curve uncertainty and demonstrated that the error in the river flow data was too large to ignore. The stage-discharge relationship can vary over time and is dependent on changes in the river geometry or roughness. In general, the curves are not suitable for ice-covered conditions and are not reliable in areas subject to unsteady flow

or changing backwater conditions. In tidal rivers, the highly complex relationship between the stage and discharge prevents a functional conformity from being deduced. Besides, rating curves (RCs) are labor-intensive and time consuming to establish.

[4] To improve discharge measurements, various techniques and instruments have been developed. For example, for the continuous measurement of water discharge, several different pieces of equipment are available, such as acoustic/ultrasonic velocity meters (AVMs/UVMs) [*Falvey*, 1983; *Laenen and Smith*, 1983; *Sloat and Gain*, 1995; *Simpson and Bland*, 2000; *Ruhl and DeRose*, 2004] and horizontal acoustic Doppler current profilers (H-ADCPs) [*Le Coz et al.*, 2008; *Nihei and Kimizu*, 2008; *Hoitink et al.*, 2009; *Sassi et al.*, 2011].

[5] AVMs/UVMs, which are based on the time-of-travel method, have the potential to produce accurate measurements of the mean line velocity along paths that diagonally transverse a channel. In AVMs, it is assumed that the acoustic path is essentially a straight line and that an accurate correlation exists between the integrated line velocity and cross-sectional average velocity. However, refractions of the acoustic path under stratified conditions cause unacceptable uncertainties in determination of the flowrate [*Falvey*, 1983; *Laenen and Smith*, 1983]. In addition, the traditional detection procedure of the arrival time, which considers the leading edge of the received signal, requires a high signal-to-noise ratio.

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[6] H-ADCPs can measure a horizontal velocity profile over a range with sufficiently strong acoustical backscatter, and beam separation problems are not encountered. The attenuated acoustical signal in flood events or turbid waters leads to inadequate results, because the horizontal profile range of H-ADCPs rapidly decreases with an increase in the suspended sediment concentration (SSC). In shallow rivers, main and sidelobe beam interference due to surface or bottom reflection contaminates the data. Moreover, H-ADCPs do not operate satisfactorily in estuaries with saltwater intrusions because the sound inflection is similar to that of AVMs. In most situations, the methods proposed by *Hoitink et al.* [2010] and *Sassi et al.* [2011] can be used to accurately estimate the discharge from velocity data in the restricted range of H-ADCPs. However, the accuracy of the estimate decreases in complex flow structures such as estuarine flows. Moreover, their method requires several hydraulic parameters, which must be deduced from labor-intensive moving boat ADCP measurements.

[7] Several techniques have been developed to estimate the flowrate from velocity data and a given bathymetry profile. In the index velocity method (IVM), the gauged section-averaged velocity is linearly regressed with AVM/H-ADCP data. In the velocity profile method (VPM), the total discharge is computed from theoretical vertical velocity profiles, which are made using H-ADCP velocity measurements and a boundary layer model. Models of velocity distribution in a channel cross section have been presented to estimate discharge [*Chen and Chiu*, 2002; *Chiu et al.*, 2005a; *Chiu et al.*, 2005b; *Chiu and Hsu*, 2006; *Givehchi et al.*, 2009; *Morse et al.*, 2010]. Chiu and coworkers implemented a series of experiments by deploying an ADCP in the velocity maxima of a natural and artificial channel. The authors hypothesized that the ratio of the mean to the maximum velocity in a cross section remains constant over the time. As a result, the discharge can be deduced from peak velocity observations. The main disadvantage of the aforementioned model is that the velocity maximum is difficult to identify, particularly in complex cross sections. Furthermore, the number of velocity sample points in cross section of a stream is often insufficient for cross-sectional average velocity estimates. Obviously, such models cannot be applied to estuarine zones.

[8] In contrast, if there is sufficient water depth and backscatter, velocity data can be collected from most parts of the cross section of a stream using moving boat ADCP measurements. However, the duration of measurement is restricted because the method is laborious. In addition, the moving boat ADCP measurements can be dangerous during extreme hydrological events such as a flood.

[9] The following situations present extreme difficulties for discharge measurement: (1) complex flows such as tidal estuaries with saltwater intrusions, (2) shallow and wide river flows, (3) fast flows with high SSC during flood events. Acoustic tomography systems have been successfully used to measure the water temperature and velocity field in coastal seas [*Kaneko and Zheng*, 1996; *Yamaoka et al.*, 2002]. Despite the periodic intrusion of a salt wedge into the channel, *Kawanisi et al.* [2010] have conducted long-term measurements on the cross-sectional average velocity and streamflow in a tidal channel using Fluvial Acoustic Tomography System (FATS), an innovative acoustic

system. In the present study, FATS was employed in a shallow, wide gravel bed river (water depth ≈ 0.6 m under low-flow conditions, width ≈ 115 m) and in the presence of an unsteady flushing flow from a nearby reservoir.

2. Overview of Fluvial Acoustic Tomography System

[10] The basic principle of FATS is similar to that of acoustic velocity meters (AVMs). In other words, the cross-sectional averaged velocity is calculated using the time-of-travel method [*Laenen and Smith*, 1983]. The AVM measures an average velocity along a straight transverse line. Therefore, several subsidiary processes must be applied to compute the discharge, including index velocity method. In contrast to AVMs, FATS sound rays scatter throughout the cross section, preventing the application of complicated post processing procedures. In unidirectional river flows, measuring only one component of the velocity would be enough to perform reliable flowrate calculations. In this case, FATS can be operated with only a couple of transducers. Otherwise, a multistation system with crossing transmission lines is required (e.g., for meandering rivers), as conducted in a coastal sea by *Yamaoka et al.* [2002].

[11] The water surface and riverbed can be considered guides for the paths of multirays. At a sufficient low-frequency and small grazing angle, the water surface can act as a nearly perfect reflector of sound [*Falvey*, 1983]. The criterion for a smooth surface is given by $\lambda > 8W \sin \varphi$, where λ is the wavelength of the acoustic signal, W is the wave height roughness (one-half of the crest-trough distance), and φ is the grazing angle between the acoustic path and water surface. The wavelength λ of the present study was approximately 57 mm. Therefore, the water surface can be considered a perfect reflector for $W < 408$ mm and $\varphi = 1^\circ$. Because the criterion is satisfied in most situations, the water surface can be considered a perfect reflector. The reflection of the acoustic signal from the riverbed is more complicated than that of the water surface. Due to the variable acoustic properties of the bed (mud to rock) and the presence of layers, accurate predictions of reflection loss are nearly impossible. However, reflections from the bed will experience more attenuation than those from the water surface. Therefore, using traditional AVMs/UVMs, the arrival time cannot be accurately detected from the leading edge of the weakly received signal [*Falvey*, 1983].

[12] Let us consider two acoustic stations, placed at horizontal spacing L in a fluid medium moving with velocity u_m . The travel times t_1 and t_2 for the forward and reverse directions are calculated by the following formulae

$$t_1 = \frac{L}{c_m + u_m} \quad (1)$$

and

$$t_2 = \frac{L}{c_m - u_m}, \quad (2)$$

respectively. From equations (1) and (2) the sound speed c_m and water velocity u_m averaged along the sound path are given by

$$c_m = \frac{L}{2} \left(\frac{1}{t_1} + \frac{1}{t_2} \right) \quad (3)$$

$$u_m = \frac{L}{2} \left(\frac{1}{t_1} - \frac{1}{t_2} \right). \quad (4)$$

The velocity component in the direction of flow (v_m) is given by

$$v_m = \frac{u_m}{\cos \theta}, \quad (5)$$

where θ is the angle between the ray path and streamline.

[13] Since the error in the travel time is negligible when both systems are synchronized with a GPS clock (the accuracy of the clock is less than 0.5 μ s), by taking the total differentiation of equations (3) and (4), the relative error of c_m and u_m can be estimated, respectively,

$$\frac{\delta c_m}{c_m} = -\frac{1}{2} \frac{\delta L}{L} \left(\frac{1}{t_1} + \frac{1}{t_2} \right) \quad (6)$$

$$\frac{\delta u_m}{u_m} = -\frac{1}{2} \frac{\delta L}{L} \left(\frac{1}{t_1} - \frac{1}{t_2} \right). \quad (7)$$

As a result, the relative errors of the average sound speed ($\delta c_m/c_m$) and the average velocity ($\delta u_m/u_m$) are equated to the relative error of the length of ray path $\delta L/L$, which is usually small in rivers with large aspect ratios (width/depth).

[14] The velocity resolution of FATS can be expressed as

$$u_r = \frac{c_m^2}{2L} \frac{1}{2f}, \quad (8)$$

where f is the central frequency of the broadband transducer. Figure 1 presents the velocity resolution as a function of L and f when $c_m = 1436 \text{ m s}^{-1}$. The velocity resolution (0.068 m s^{-1}) in the present study ($L = 301.96 \text{ m}$, $f = 25 \text{ kHz}$) is shown in Figure 1. This relatively low-velocity resolution signals the necessity of averaging the results over a certain interval. The statistical reliability of the FATS increases with the square root of the number of the samples per ensemble. To reduce the uncertainty of the

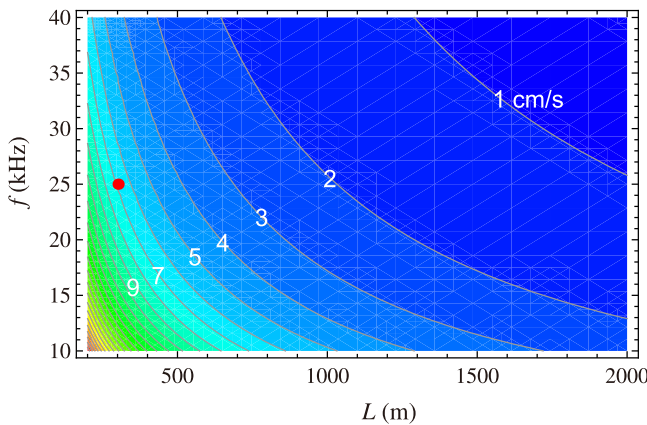


Figure 1. Velocity resolution of FATS as a function of the ray path length and acoustic frequency of the carrier signal.

FATS, 40 samples were averaged in the present study. Therefore, the uncertainty due to the low-velocity resolution can be decreased to $0.068/\sqrt{40} = 0.011 \text{ m s}^{-1}$.

[15] Using FATS, the flowrate is calculated as follows:

$$Q = A(H) v_m \sin \theta = A(H) u_m \tan \theta, \quad (9)$$

where A is the cross-sectional area of the sound path and is a function of the water level H .

[16] To determine the arrival time of traveling sound mixed with noises correctly, the transmitted signal is phase-modulated by applying the pseudo-random M-sequence [Simon *et al.*, 1985; Zheng *et al.*, 1998], i.e., the carrier signal is phase-modulated by taking the product of the M-sequence. By transmitting the modulated signal within a broadband frequency range, the signal-to-noise ratio (SNR) increases according to the relation $(2^n - 1)$, where n denotes the order of the M-sequence. In this study, a tenth-order M-sequence was employed. By taking the cross correlation between the received signal and the original phase-modulated signal it becomes possible to determine the arrival time with very high accuracy. Note that GPS clock accuracy plays a key role in this process. As a result, long-term measurements of the cross-sectional average velocity and river discharge were successfully obtained, despite the strong tidal effects described in our previous work [Kawanisi *et al.*, 2010].

[17] To precisely estimate the cross-sectional-averaged velocity (v_m), the ray paths must cover the cross section as much as possible. The sound rays of FATS are likely to cover the whole cross section in a freshwater environment, i.e., in the absence of stratification caused by seawater intrusions [Kawanisi *et al.*, 2010]. In contrast to AVMs/UVMs that are only resolve the leading edge of received signal, FATS is capable to detect the arrival time of a group of rays transmitted at each measurement, i.e., the arrival time was detected through the cross-correlation wave between the received signal and the M-sequence.

[18] Figure 2 shows several views of the state-of-the-art FATS developed at Hiroshima University. The broadband transducer (Model T257, Neptune Sonar Ltd., see <http://www.neptune-sonar.co.uk/wp/index.php/transducer-products/transducers/>) is 55 mm in diameter, and the horizontal and

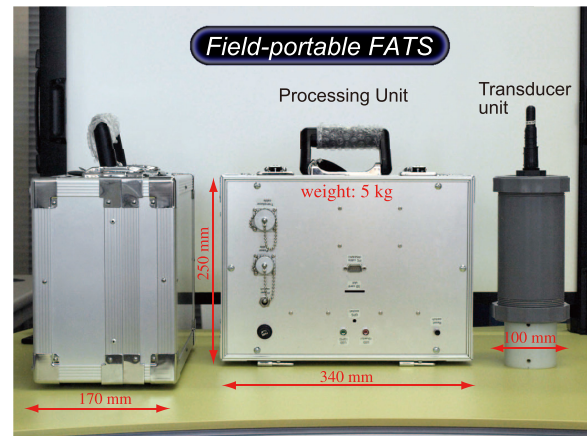


Figure 2. Fluvial Acoustic Tomography System (FATS).

vertical beam patterns of the transducer are omnidirectional and hemispherical, respectively. The bandwidth is approximately 20 kHz and the source level is 197 decibels. FATS is small in size, lightweight (5 kg) and has low power consumption (4 W). Thus, unlike AVMs/UVMs, FATS does not require extensive infrastructure. Moreover, due to the small broadband transducer with omnidirectional/hemispherical beam patterns and multiple ray methods, the flowrate can be measured under the rough hydraulic conditions that dominate shallow rivers or estuarine zones with saltwater intrusions.

3. Field Site and Methodology

3.1. Study Area

[19] Field investigations were performed in the Gono River, Miyoshi City, western Japan (Figure 3), and a 115 m wide straight reach of the river was selected for the location of the transducers. The bed slope around the observation site was 0.11‰ and the Manning roughness estimated from the water surface profile was approximately 0.03. The water depth was approximately 0.6 m under low-flow conditions.

[20] The two major tributaries of the Gono River are united at 2.7 km upstream from the observation site, and the confluent rivers meet the Gono River 0.9 km downstream from the junction. The Gono River catchment area is 3900 km² and the annual mean discharge at the Ozekiyama

gauging station (Figure 3a), which is located 1.1 km upstream of the observation site, is estimated to be 73 m³ s⁻¹.

[21] Figure 4 shows the grain size distribution of the bed sediment. The grain size distribution presented an average median size (D_{50}) of 27 mm and an average D_{90} of 115 mm. The grain size distribution indicated a deficiency of fine sand, silt and clay.

[22] The Gono River is a perfect place to catch Ayu sweetfish. The Ayu feeds on sphagnum growing on bedrock in streams. To encourage the growth of new sphagnum, water from the Haizuka Reservoir, which is built on one of the Basen River tributaries, is discharged once a year. The Haizuka Reservoir is located 26 km upstream from the observation site.

[23] The properties of the water were regularly measured at the center of the river near the observation site, as shown in Figure 3a. The hourly data measured at the water quality observatory are available on the MLIT website.

3.2. FATS Experiments

[24] Continuous measurements of the flowrate were carried out on 21–24 January, 22–24 March, and 31 October to 13 December 2011. The second measurement was conducted during a dam flushing. In the first and third measurements, reference discharge data were collected by performing moving boat ADCP measurements. The ADCP

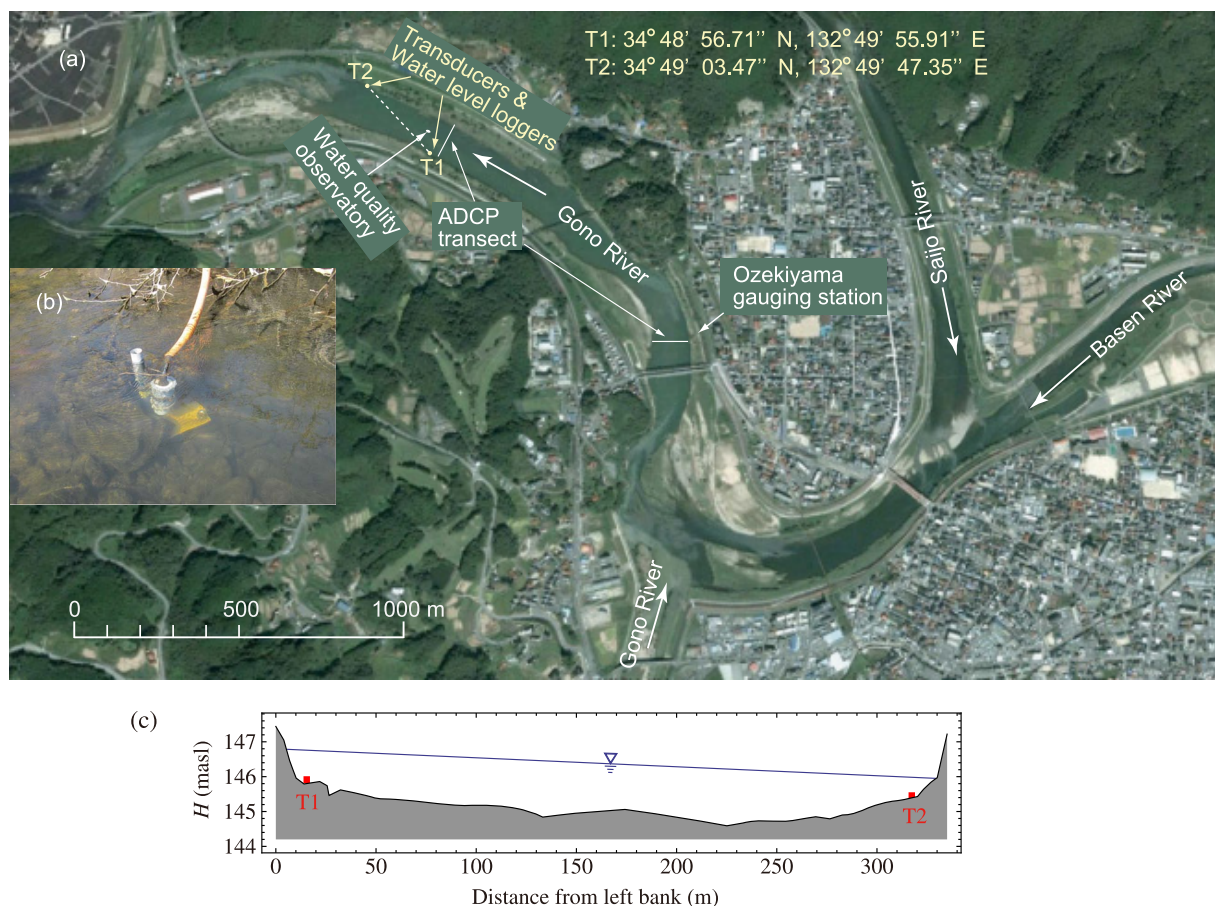


Figure 3. Study area and experimental site: (a) location map showing the position of the transducers, ADCP transects, gauging station and water quality observatory; (b) view of the transducer unit installed at the right bank; (c) oblique cross-section in which sound rays of FATS create multipath.

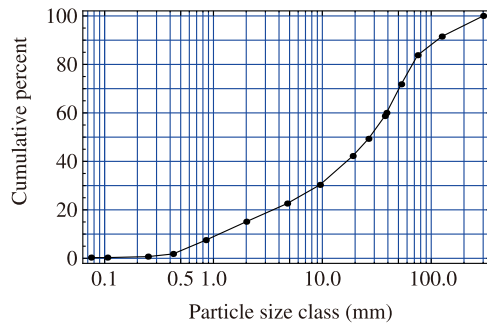


Figure 4. Grain size distribution of bed sediment.

transect relative to the FATS acoustic path can be found in Figure 3a. A couple of broadband transducers (T257, Neptune Sonar Ltd.) were installed diagonally across the river, as shown in Figure 3a. A view of the transducer (T2) on the right bank is shown in Figure 3b. Figure 3c presents the oblique cross section, in which the sound rays of FATS make up a multipath. The bottom topography is given by surveying several points of the bed every 8 m using a total station. The horizontal and vertical accuracy of measurements was less than 0.006 m and 0.03 m, respectively.

[25] The central frequency of the transducers was set to 25 kHz, the angle between the acoustic pass and stream direction θ in the ADCP velocity measurements was equal to 22 degrees, the transducers were mounted at the height of 0.3 m above the bed. The elevation of the left and right transducers was 145.91 m above mean sea level (masl) and 145.48 masl, respectively (Figure 3c). Horizontal distance between the two transducers was 301.96 m. The acoustic pulses of FATS were triggered by a GPS clock and were transmitted simultaneously from both transducers every minute.

[26] The water levels were measured every minute by water level loggers attached to the both transducers. The water level loggers also measured the pressure and temperature, and a barometer on the riverbank was used to record atmospheric pressure. The accuracy of the water level sensors and the barometer were ± 5 mm and ± 3 mm, respectively.

3.3. ADCP Measurement Campaigns

[27] To establish a set of reference discharge data for evaluating FATS, 11 transects were made in the first and third field tests by a Teledyne RDI StreamPro ADCP operating at 2.0 MHz (see http://www.rdiinstruments.com/data-sheets/streampro_ds_lr.pdf). Each transect extended from the immediate left shore and ended 8 m from the right shore (the edge type was set to rectangular).

[28] A sample transect is plotted in Figure 3a. The site conditions allowed the default settings of StreamPro to be applied. Namely, the depth of the cell size was 0.05 m, and the range to the first depth of the center of the cell was 0.19 m, assuming that the sensor head was 0.05 m below the water surface. Water mode 12 was selected, and 6 pings per ensemble were employed. The flow was steady during the tests period and the float speed was maintained at approximately 0.2 m s^{-1} . The relative velocity between the ADCP and the bottom of the stream, which was used to measure the water velocity, was estimated by ADCP bottom tracking. Because average size of the bed constituent material is large

(Figure 4), drift in the bottom tracking due to moving bed effects was not observed. The bottom tracking data were also used to derive the position of the ADCP with respect to the starting point of each transect.

4. Results

4.1. Discharge Validation

[29] The flow rates estimated by FATS were validated by moving boat ADCP campaigns. Besides, FATS estimates were compared to those from latest rating curves (RCs) available, which were established from stream gauges at the Ozekiyama gauging station through 2010. Only 2% of the FATS data was rejected due to the low SNR which does not have any implication to the coherency of records considering the sampling frequency of 1/60 Hz. The threshold level of the SNR was set to 11 decibels.

[30] Figure 5 presents the results of stream gauges at the Ozekiyama gauging station in 2010. The ADCP transect is shown in Figure 3a. As shown in Figure 5b, the uncertainty of RC method increases by 10% at low flows. For the larger discharges ($Q \geq 50 \text{ m}^3 \text{ s}^{-1}$), the discharge uncertainty is approximately $\pm 5\%$. The root-mean-square of residual (RMSR) between the RC and ADCP estimates is $2.44 \text{ m}^3 \text{ s}^{-1}$. The RC denoted by green line was used to compare the FATS estimates because the 2011 RC is not established yet.

[31] Figure 6 presents a screenshot from WinRiver II software (Teledyne RD Instruments), showing the velocity distribution in the first transect on 21 January 2011. Figure 6b shows a typical horizontal view of the depth-averaged velocities along the ADCP track. Accordingly, the river flow at the site can be considered unidirectional. The ADCP discharges were extracted from WinRiver II software. For each ensemble bad bins portion did not exceed 2%, and bad ensembles in each transect did not exceed 3.5%. The contribution of the

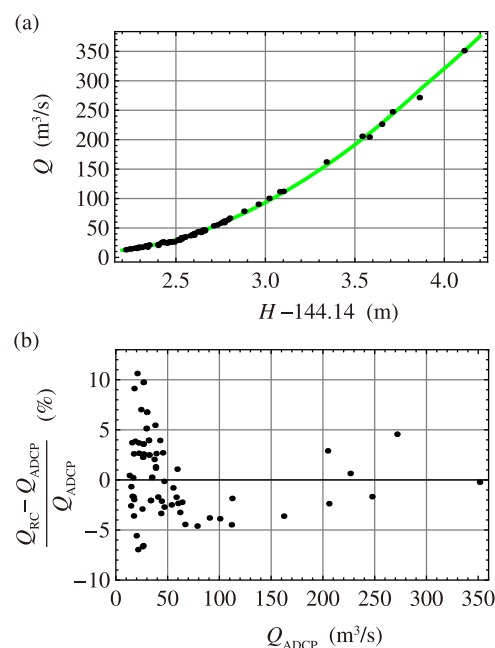


Figure 5. (a) Rating curve (green line) at the Ozekiyama gauging station in 2010; (b) Relative difference between RC and ADCP estimates.

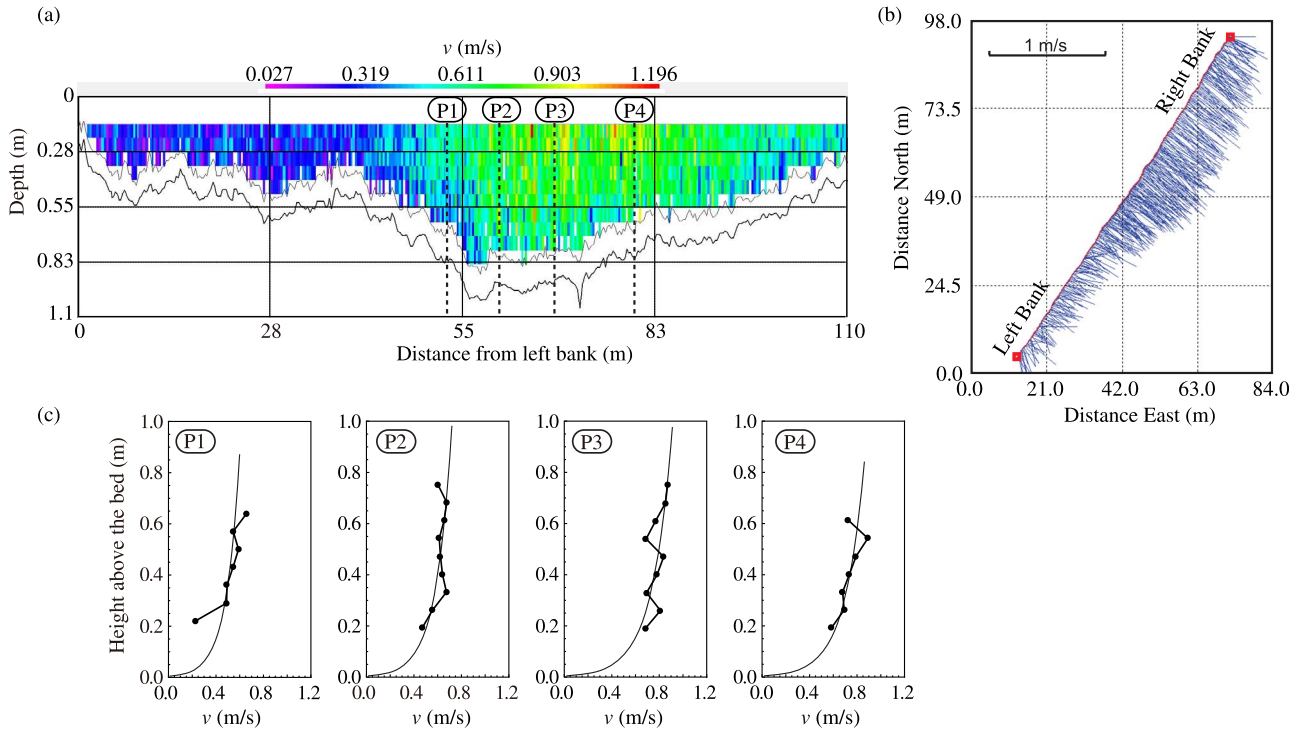


Figure 6. Velocity distribution obtained by moving-boat ADCP measurement: (a) velocity distribution in the cross-section; (b) velocity vectors on the transect; (c) typical vertical profiles of the ADCP velocity at P1-P4 in Figure 6a; solid lines represent the 1/4 power law.

unmeasured near-shore areas (too shallow for measurement) was less than 1% of total discharge. As a result, unmeasured near-shore discharge did not significantly bias the ADCP discharge estimates. However, between the surface and the first ADCP cell, a region with a width of 0.19 m was not measured. In addition, there remains another unmeasured area near the bottom of the profile occurred due to sidelobe interference.

[32] The general form of the power law is expressed as

$$\frac{u(z)}{u_{\max}} = \left(\frac{z}{h}\right)^{1/m} \quad (10)$$

where h is the water depth, $u_{\max}$ the velocity at $z = h$. The power $1/m$ can be related to the Manning roughness n as follows:

$$\frac{1}{m} = \frac{\sqrt{g}n}{\kappa h^{1/6}}, \quad (11)$$

where κ is the Karman constant. Substituting $n = 0.03$ and $h = 1$ m, $1/m \approx 1/4$ is obtained.

[33] Thus, unmeasured areas near the top and bottom of the vertical profile were estimated by fitting measured data to a [1/4] power curve, which was used for extrapolation. The contribution of top (bottom) discharge to the total discharge was 21.9% (17.6%). Figure 6c displays some examples of the vertical distributions of ADCP longitudinal velocity. The solid line denotes a [1/4] power curve. The determination coefficients R^2 were greater than 0.9. Thus, the [1/4] power extrapolation was relevant to this site.

[34] Figures 7a and 8b shows the temporal variations in the flowrate according to FATS, moving boat ADCP, and the RC. The variation range of water level was about 1 m during the last field test (Figure 8a). Evidently, the FATS estimates are in good agreement with the ADCP estimates. The relation between the FATS and ADCP discharges is shown in Figure 9. The relative difference between ADCP and FATS estimates is approximately within $\pm 10\%$ range.

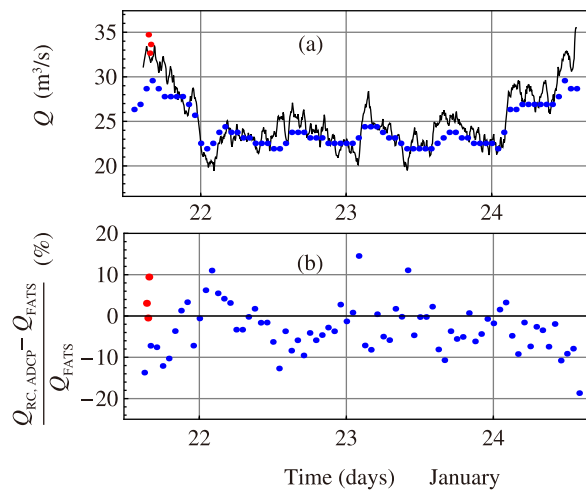


Figure 7. (a) Temporal variations in flow rates measured by FATS (black line), moving boat ADCP (red disks) and rating curve method (blue disks) at low-flow condition. (b) Temporal variations of relative flowrate differences.

The RMSR between FATS and ADCP is $2.41 \text{ m}^3 \text{ s}^{-1}$, that indicates the solidity of the FATS measurements.

[35] From the discrepancy between the RC and FATS/ADCP results particularly during low-flow conditions, one may infer that RC estimates are reliable only in large discharges, saying $Q \geq 40 \text{ m}^3 \text{ s}^{-1}$. Apparently, in low flows (31 October to 12 November) and also in extreme events RC method may overestimate the discharge by 50% (Figure 8b). The RC made from stream gauges in 2010 may be biased in low flows in 2011 mainly due to the morphological changes in riverbed. Figure 7b gives evidence that the uncertainty of RC method is roughly $\pm 15\%$ on 21–24 January and a larger RMSD of $5.80 \text{ m}^3 \text{ s}^{-1}$ in the low flow.

4.2. Continuous Observation During Dam Flushing

[36] The water from the Haizuka Reservoir was discharged from 08:00 UTC on 23 March to 08:00 UTC on 24 March 2011. Figure 10a presents the discharge hydrograph. The maximum flowrate was $100 \text{ m}^3 \text{ s}^{-1}$ and total discharge volume was $2.31 \times 10^6 \text{ m}^3$. The discharge was low (approximately $3 \text{ m}^3 \text{ s}^{-1}$) from 20:10 on 23 to 08:00 on 24. Thus, the discharge volume from 08:00 to 20:10 on 23, $2.18 \times 10^6 \text{ m}^3$, was compared to the discharge volume estimated by FATS.

[37] FATS measurements were performed from 12:15 on 22 to 15:14 on 24 March 2011. Unfortunately, moving boat ADCP campaigns were difficult to perform because the expected arrival time of the flashing discharge was during the middle of the night. The mean water depth of the section, in which sound rays traveled, ranged from 0.85 m to 1.3 m during the observation period.

[38] Figure 10b shows the hydrograph estimated by FATS. Approximately 5% of the data obtained at 1 min intervals had insufficient SN ratios ($< 11 \text{ dB}$). The FATS estimates are in good agreement with the hourly flow rates estimated from the rating curves of the Ozekiyama gauging station. As shown in Figure 10c, the relative difference between the RC and FATS estimates are within $\pm 10\%$ and RMSR is $3.77 \text{ m}^3 \text{ s}^{-1}$. The flushing discharge volume estimated from the hump in the hydrograph is $2.18 \times 10^6 \text{ m}^3$. The evaluation period of the flushing discharge volume is represented by the arrows shown in Figure 10b. The discharge volume of $2.18 \times 10^6 \text{ m}^3$ is equal to the volume of

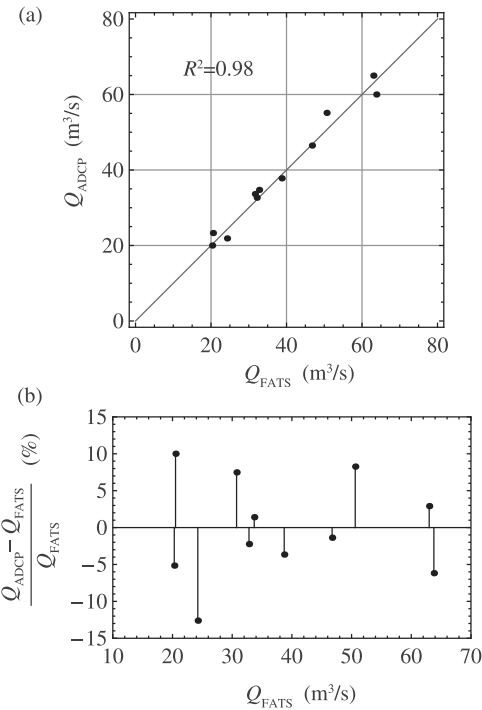


Figure 9. (a) Relation between FATS and ADCP estimates; (b) Relative flowrate differences.

water discharged from the Haizuka Reservoir, suggesting that FATS can successfully measure the flushing discharge.

[39] The cross-sectional average temperature can be estimated from the sound speed measured by FATS and the water depth. Various equations have been proposed to predict the acoustic velocity in water. *Medwin* [1975] proposed a formula for the sound speed c as a function of the temperature T ($^{\circ}\text{C}$), salinity S , and depth D (m) at $0 \leq T \leq 35^{\circ}\text{C}$, $0 \leq S \leq 45$, and $0 \leq D \leq 1000 \text{ m}$:

$$c = 1449.2 + 4.6T - 0.055T^2 + 2.9 \times 10^{-4}T^3 + (1.34 - 0.01T)(S - 35) + 0.016D. \quad (12)$$

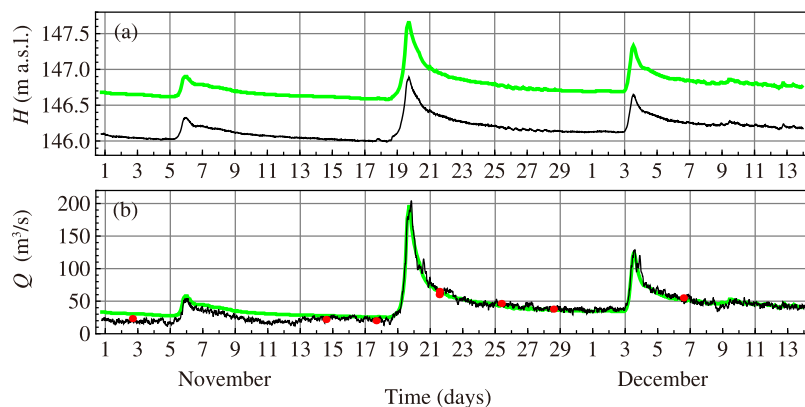


Figure 8. Temporal variations in (a) water elevations at the Ozekiyama gauging station (green line) and FATS observation site (black line); (b) flow rates by RC (green line), FATS (black line), and moving boat ADCP (red disks).

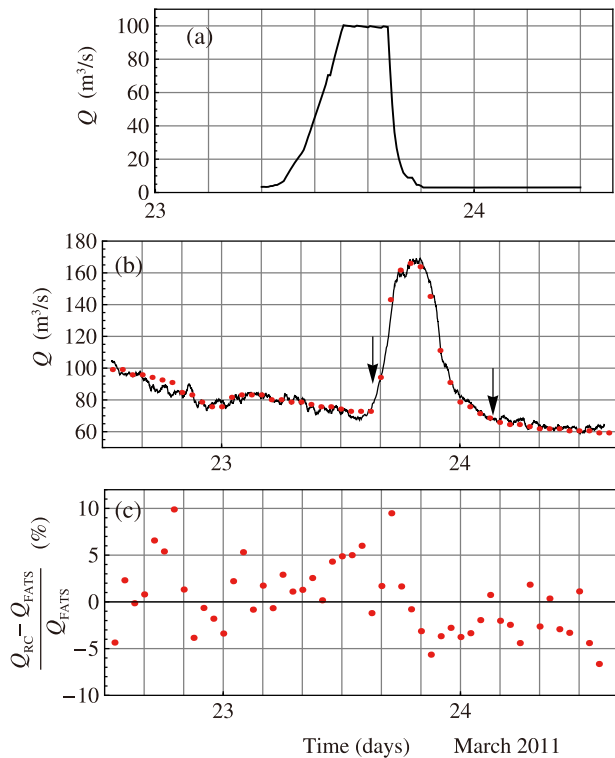


Figure 10. Temporal variations in flow rates (a) from the Haizuka reservoir; (b) measured by FATS (solid line) and rating curve method (red disks). Arrows indicate start and end times for evaluation of flushing discharge volume. (c) Temporal variation in relative difference on flowrate.

[40] Figure 11a compares the temporal variation in the temperature from FATS, the temperature sensors of the water level loggers, and the water quality observatory located at the center of the river. The temperature derived from the sound speed is very close to the values obtained

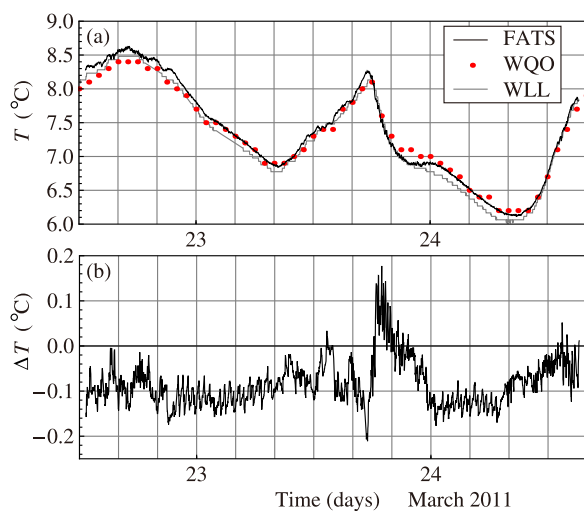


Figure 11. Temporal variations in the water temperature: (a) temperature data from FATS, the water quality observatory (WQO) and water level loggers (WLL); (b) temperature difference between T_{WLL} and T_{FATS} .

from the temperature sensors. The temperature resolution of FATS can be expressed as:

$$T_r = \frac{c^2}{2\alpha f L}, \quad (13)$$

where $\alpha = 4.6 - 0.11 T + 8.7 \times 10^{-4} T^2 - 0.01(S - 35)$. Assuming $T = 7.5^\circ\text{C}$ and $S = 0$ the equation (13) yields $T_r = 0.0327^\circ\text{C}$, which implies that the resolution of FATS for the measurement of the mean temperature is high. The resolution and accuracy of the temperature sensor in the water level logger was 0.1°C and 0.37°C , respectively.

[41] As shown in Figure 11b, the difference between the mean temperature of the two water level loggers and FATS is approximately $\pm 0.2^\circ\text{C}$.

[42] The sound attenuation due to the presence of suspended solids can reduce the SNR of FATS. Figure 12 shows the temporal variation of the SNR and turbidity at the water quality observatory (the turbidity data was available by 31 March 2011). The turbidity was difficult to convert to SSC because the turbidity meter was not calibrated against water samples and the particle size distribution of suspended sediment is not available.

[43] The SNR of FATS gradually decreases to 13.5 dB as the turbidity increases tenfold by the flushing. Comparing the two panels of Figure 12 suggests a time lag between SNR dip and turbidity maxima that may stem from grain size effects of suspended sediment. The flow depth and SNR appear to be directly related in extremely shallow (mean depth < 0.9 m) streams. Reflection of sound waves increases considerably in extremely shallow streams. Because reflections from the bed experience more attenuation, SNR decreases under extremely shallow conditions.

5. Discussion

[44] When large fluctuations are observed in the direction of the stream, a four-station system with two crossing

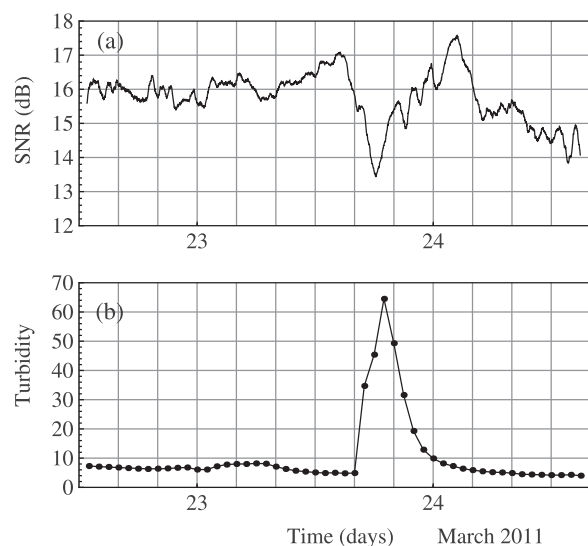


Figure 12. Temporal variations in (a) SN ratio of FATS and (b) turbidity at OWQ.

transmission lines must be used to detect the stream direction [Laenen and Smith, 1983]. The effect of fluctuations in the stream direction on the flowrate can be estimated from using the following equation:

$$\frac{\delta Q}{Q} \simeq \frac{\delta \theta}{\cos \theta \sin \theta} + \frac{2 \delta \theta^2}{\cos^2 \theta}. \quad (14)$$

Figure 13 shows the relative error in the flowrate as a function of θ and $\delta \theta$. The optimal angle for fluctuations in the direction of the stream is approximately 45° . In the present case, $\theta = 22^\circ$, and fluctuations of $\pm 1^\circ$ introduce relative discharge errors of -5.0 and 5.1% . The standard deviation of the cross-sectional average stream direction estimated by the ADCP transects was approximately 1.5° . Therefore, fluctuations in the stream direction can potentially induce considerable errors in FATS discharge measurements. In the next phase, to minimize such errors discharge measurements by FATS with two crossing transmission lines are on the menu.

[45] Water temperature deduced from Medwin's formula readily changes as a function of the path length. Because the influence of depth is negligible, the error in the temperature due to changes in the path length can be expressed as follows:

$$\delta T = c \frac{\partial T}{\partial c} \frac{\delta L}{L} \approx 338 \frac{\delta L}{L}. \quad (15)$$

Therefore, the relative error of the path length leads to large temperature errors. If it is accepted that the temperature deviance ($\pm 0.2^\circ\text{C}$, Figure 11(b)) is produced only by variations/fluctuations in the path length, the relative error of the path length is:

$$\frac{\delta L}{L} = \frac{\delta T}{c \partial T / \partial c} \approx \pm 0.06\%. \quad (16)$$

As a result, one can infer that the error in the velocity of FATS due to changes in the path length is negligible, because the relative error of the average velocity ($\delta u_m / u_m$) is equal to $\delta L / L$. Thus, most of the uncertainty and error in the velocity measurements of FATS can be attributed to the

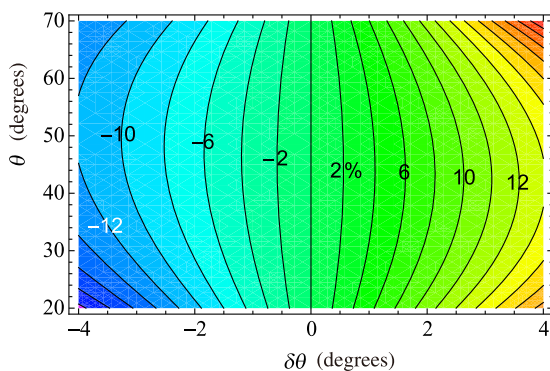


Figure 13. Relative error of flowrate due to stream direction error.

fluctuations of stream direction, acoustic artifacts and low-velocity resolution.

6. Conclusions

[46] Continuous measurements of the flowrate were conducted in an extremely shallow and wide gravel bed river using *Fluvial Acoustic Tomography System* (FATS) a state-of-the-art acoustic system. A FATS equipped with a couple of 25 kHz broadband transducers with horizontally omnidirectional and vertically hemispherical beam patterns can be used to estimate the cross-sectional average velocity from multiple ray paths that cover the cross section of a stream, even though FATS is based on the time-of-travel method used in traditional AVMs/UVMs.

[47] The water depth of the gravel bed river (Gono River, Japan) was approximately 0.6 m and the river width was 115 m under low-flow conditions. The flow rates measured by FATS were compared to those of moving boat ADCP over a range of 20 to 65 $\text{m}^3 \text{s}^{-1}$. Although several areas in the cross section could not be directly measured by ADCP, the ADCP discharges, which were calculated by WinRiver II software (Teledyne RD Instruments), were in good agreement with the FATS discharges (the relative differences between ADCP and FATS estimates approximately were within $\pm 10\%$). Although the flow rates by RC method are poorly rated at low flows ($Q < 40 \text{ m}^3 \text{s}^{-1}$), the RC probably gives reliable discharge estimates under larger flow conditions ($Q \geq 40 \text{ m}^3 \text{s}^{-1}$).

[48] Another deployment of FATS was performed during reservoir flushing. The mean water depth of the section in which sound rays traveled ranged from 0.85 m to 1.3 m during the observation period. The FATS discharges were in good agreement with the rating curve method results (the relative difference between RC and FATS estimates was within $\pm 10\%$). Suspended particles due to reservoir flushing decreased the SNR of FATS by around 13.5 dB. Approximately 5% of the data obtained at 1 min intervals presented insufficient SNR ($< 11 \text{ dB}$).

[49] The water temperature deduced from the FATS sound speed data closely tracked the temperature sensor values. The difference between the temperature data of FATS and the temperature sensors was $\pm 0.2^\circ\text{C}$. If the temperature difference is due to changes in the ray path length, the relative error of the path length and velocity is $\pm 0.06\%$. Therefore, fluctuations in the path length do not significantly affect the discharge estimates.

[50] The present study and previous work [Kawanisi et al., 2010] strongly suggest that FATS is a reliable and accurate technique for the long-term measurement of river discharge, even in shallow, wide rivers or tidal estuaries with saltwater intrusions. Thus, FATS is a prospective acoustic system for the continuous monitoring of river discharge and temperature/salinity variations.

[51] The fluctuations in the stream direction can induce considerable errors in FATS discharge measurements. In the next step, to promote the measurements quality crossing transmission lines will be used.

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